

Runxi Cheng 程润曦

Master's Student in Artificial Intelligence

TSINGHUA UNIVERSITY · BEIJING, CHINA

nathyyy1120@gmail.com

scholar.google.com/citations?user=5DsuoHwAAAAJ

github.com/nathanielyvo

dblp.org/pid/385/1319

nathanielyvo.github.io



I AM ADVISED BY Prof. Chun Yuan at Tsinghua University. I received my Bachelor's degree in Information Management and Information Systems from Beihang University in 2022, and have been a research intern at Microsoft Research Asia (Artificial Intelligence Reasoning Group), mentored by Yeyun Gong and Weizhu Chen, and at Tencent IEG (Basic Research Group).

My research aims to make large language models *more capable and more efficient to build*. I work on **scalable LLM pre-training and architectures** — mixture-of-experts and conditional memory — on **model merging and knowledge transfer**, including data-free merging with task vectors and knowledge distillation, and on **reasoning** through self-training and data synthesis.

PUBLICATIONS

* *Equal contribution*

Conference Papers

- [1] **Memory Grafting: Scaling Language Model Pre-training via Offline Conditional Memory**
Runxi Cheng, Yuchen Guan, Yongxian Wei, Qianpu Sun, Qixiu Li, Sinan Du, Feng Xiong, Chun Yuan, Yan Lu, Yeyun Gong
Conference on Neural Information Processing Systems (NeurIPS) 2026
TL;DR Stores a frozen pretrained model's offline hidden states for frequent n-grams as lookup memory, improving LLM pre-training over MoE and Engram baselines with limited overhead.
- [2] **Whoever Started the Interference Should End It: Guiding Data-Free Model Merging via Task Vectors**
Runxi Cheng*, Feng Xiong*, Yongxian Wei, Wanyun Zhu, Chun Yuan
International Conference on Machine Learning (ICML) 2025
TL;DR Shows that a linear layer's task vectors lie approximately in the subspace spanned by its inputs, and uses them to eliminate merging interference without any data — +10.9% over data-free baselines.
- [3] **Enhancing Logits Distillation with Plug&Play Kendall's τ Ranking Loss**
Yuchen Guan*, **Runxi Cheng***, Kang Liu, Chun Yuan
International Conference on Machine Learning (ICML) 2025
TL;DR A plug-and-play Kendall's τ ranking loss complements KL-based logit distillation by rebalancing gradients toward low-probability classes, consistently improving students on CIFAR-100, ImageNet and COCO.
- [4] **Spectral Characterization and Mitigation of Sequential Knowledge Editing Collapse**
Chi Zhang, Mengqi Zhang, Xiaotian Ye, **Runxi Cheng**, Zisheng Zhou, Ying Zhou, Pengjie Ren, Zhumin Chen
Annual Meeting of the Association for Computational Linguistics (ACL) 2026, Main Conference (Long Papers)
TL;DR Shows sequential edits collapse LLMs by disrupting dominant singular directions of pretrained weights, and proposes REVIVE, a plug-and-play filter that preserves that subspace across up to 20,000 edits.
- [5] **OptMerge: Unifying Multimodal LLM Capabilities and Modalities via Model Merging**
Previously titled "Unifying Multimodal Large Language Model Capabilities and Modalities via Model Merging".
Yongxian Wei, **Runxi Cheng**, Weike Jin, Enneng Yang, Li Shen, Lu Hou, Sinan Du, Chun Yuan, Xiaochun Cao, Dacheng Tao
International Conference on Learning Representations (ICLR) 2026
TL;DR The first model-merging benchmark for multimodal LLMs, plus OptMerge, a data-free method that denoises task vectors to merge expert capabilities and modalities into one model.
- [6] **HS-STaR: Hierarchical Sampling for Self-Taught Reasoners via Difficulty Estimation and Budget Reallocation**
Feng Xiong*, Hongling Xu*, Yifei Wang, **Runxi Cheng**, Yong Wang, Xiangxiang Chu
Conference on Empirical Methods in Natural Language Processing (EMNLP) 2025, Main Conference — Oral
TL;DR Estimates problem difficulty with lightweight reward-guided pre-sampling, then reallocates the sampling budget to boundary-level problems — better math self-training at no extra sampling cost.

[7] **Mixture of Neuron Experts**

Runxi Cheng, Yuchen Guan, Yucheng Ding, Qingguo Hu, Yongxian Wei, Chun Yuan, Yelong Shen, Weizhu Chen, Yeyun Gong

arXiv preprint, under review

TL;DR Decomposes MoE experts into neuron-granular experts and keeps only the top-k activated neurons in each, matching standard MoE with half of the activated MoE-layer parameters.

[8] **Multi-Task Model Merging via Adaptive Weight Disentanglement**

Feng Xiong*, **Runxi Cheng***, Wang Chen, Zhanqiu Zhang, Yiwen Guo, Chun Yuan, Ruifeng Xu

arXiv preprint, under review

TL;DR Subtracts a learned shared redundant vector so that task vectors become closer to orthogonal, reducing interference and improving multi-task model merging.

[9] **Learning to Pose Problems: Reasoning-Driven and Solver-Adaptive Data Synthesis**

Earlier arXiv versions were titled "... for Large Reasoning Models".

Yongxian Wei, Yilin Zhao, Zixuan Hu, Li Shen, Xinrui Chen, **Runxi Cheng**, Sinan Du, Hao Yu, Chun Yuan, Dian Li

arXiv preprint, under review

TL;DR Trains a problem generator that reasons about problem design before writing, and uses solver feedback as a reward to synthesize problems adapted to the solver's competence.

Other Contributions

[10] **SIGMA: An AI-Empowered Training Stack on Early-Life Hardware**

Lei Qu, Lianhai Ren, Peng Cheng, Rui Gao, Ruizhe Wang, Tianyu Chen, Xiao Liu, Xingjian Zhang, Yeyun Gong, Yifan Xiong, Yucheng Ding, Yuting Jiang, Zhenghao Lin, Zhongxin Guo, Ziyue Yang

Technical report, arXiv 2025

The Microsoft Research report acknowledges me as a contributor for technical exploration (Appendix A); I am not on the author byline.

TL;DR An open-source training stack for early-life AI accelerators: 94.45% effective cluster utilization, and a 200B MoE model trained stably on 2,048 accelerators.

RESEARCH EXPERIENCE

Microsoft Research Asia

FEB 2025 – FEB 2026

Research Intern, Artificial Intelligence Reasoning Group

Beijing

- Contributed to the design and implementation of MSRA's next-generation large language model.
- Foundational research on LLM pre-training methodology, with an emphasis on training stability and efficiency.
- Research on optimizing mixture-of-experts (MoE) architectures for parameter efficiency and inference performance.

RELATED WORK Memory Grafting [1], MoNE [7], SIGMA (*contributor*) [10]

Tencent

JUL 2024 – JAN 2025

Research Intern, IEG · Basic Research Group

Shenzhen

- Research on model merging and parameter-efficient fine-tuning of large language models.
- Completed two papers on model merging, one of which was accepted to ICML 2025.

EDUCATION

Tsinghua University

Master's in Artificial Intelligence

SEP 2023 – PRESENT

Beijing, China

Beihang University

Bachelor's in Information Management and Information Systems

SEP 2018 – JUL 2022

Beijing, China